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Motivation & Problem

Power-system operators publish enormous quantities of dispatch data (hourly generation, demand, clearing prices) **without disclosing** underlying cost curves or operating limits.

Recovering these latent parameters is central to:

- **Marginal emission factor (MEF)** forecasting
- Counterfactual policy analysis
- Calibrating capacity-expansion models
- Detecting market power

Goal: Given T demand–dispatch pairs $\{(d_t, g_t)\}_{t=1}^T$ produced by an unknown DC-OPF, recover

$$\theta^* = (f^*, g_{\max}^*, p_{\max}^*).$$

Forward DC-OPF

For incidence $A \in \mathbb{R}^{n \times m}$, the DC optimal power flow is:

$$\begin{aligned} \min_{g, p, \theta} \quad & f^\top g + \frac{\varepsilon}{2} \|g\|_2^2 + \frac{\tau}{2} \|p\|_2^2 \\ \text{s.t.} \quad & Ap + g = d \\ & p = \text{diag}(b) A^\top \theta \\ & 0 \leq g \leq g_{\max}, \quad |p| \leq p_{\max} \end{aligned}$$

Tikhonov regularizers ($\varepsilon=0.1, \tau=0.01$) make the QP **strongly convex**, so the solution map $(d, f, g_{\max}, p_{\max}) \mapsto (g^*, p^*)$ is unique and Lipschitz-differentiable.

Setting $b \equiv 1$ gives a network-flow relaxation; the physics can be tightened with real susceptances as needed.

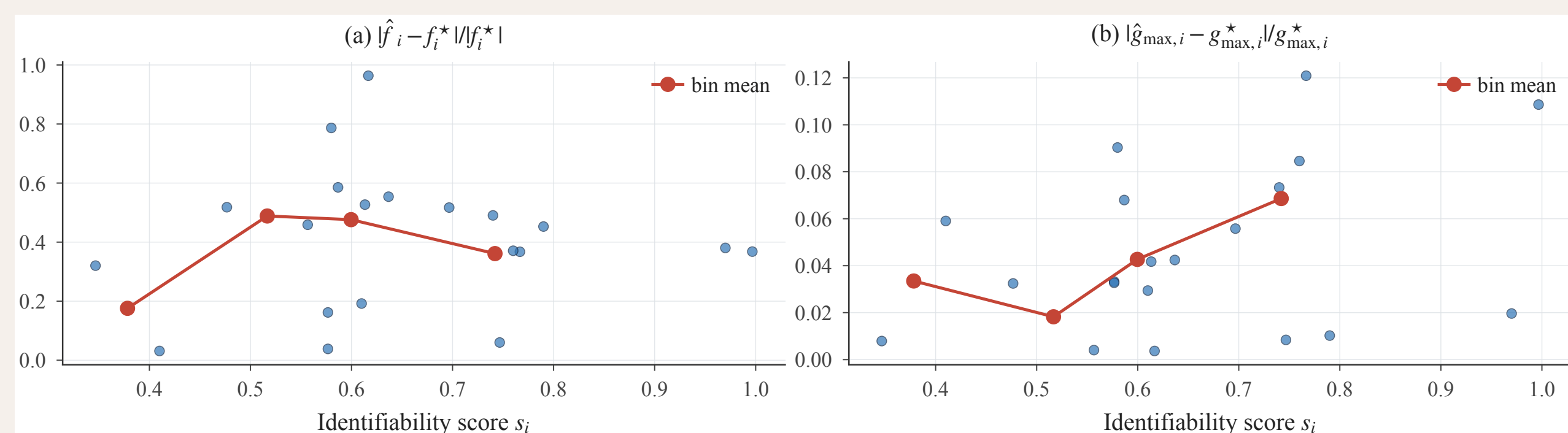
Active-Set Identifiability

A generator's cost is observable *only* when its dispatch is sensitive to demand changes.

Lemma. Fix $(d, g_{\max}, p_{\max})$. If generator i is strictly interior ($0 < g_i^* < g_{\max,i}$) and the KKT system is non-degenerate, then $\partial g_i^* / \partial f_i \neq 0$. Conversely, if generator i is at a bound for all t , then f_i is unidentifiable from $\{(d_t, g_t)\}$.

Empirical identifiability score:

$$s_i = \frac{1}{T} \sum_{t=1}^T \mathbf{1}[\text{tol} \leq g_{t,i}^* \leq g_{\max,i} - \text{tol}]$$



Per-generator recovery error vs. identifiability score. High- s_i generators (strictly interior) are reliably recovered; always-binding generators concentrate at high error.

Differentiable Inverse Estimator

We embed the forward QP in a CvxpyLayer and minimize:

$$\mathcal{L}(\hat{\theta}) = \frac{1}{T} \sum_{t=1}^T \text{Huber}(g^*(d_t; \hat{\theta}) - g_t^{\text{obs}}) + \lambda_2 \|\hat{\theta}\|_2^2 + \lambda_L \text{Lap}(\hat{F})$$

- $\hat{\theta} = (\hat{f}, \hat{g}_{\max}, \hat{p}_{\max})$ via softplus parameterization (enforces non-negativity)
- $\hat{F} \in \mathbb{R}^{S \times n}$: stratified cost table (one cost vector per hour-of-day stratum s)
- $\text{Lap}(\hat{F}) = \text{tr}(\hat{F}^\top L \hat{F})$: cycle Laplacian regularizes hour-to-hour drift
- Gradients $\nabla_{\hat{\theta}} g^*$ via implicit KKT differentiation (cvxpylayers)
- Optimizer: Adam + cosine LR decay + early stopping

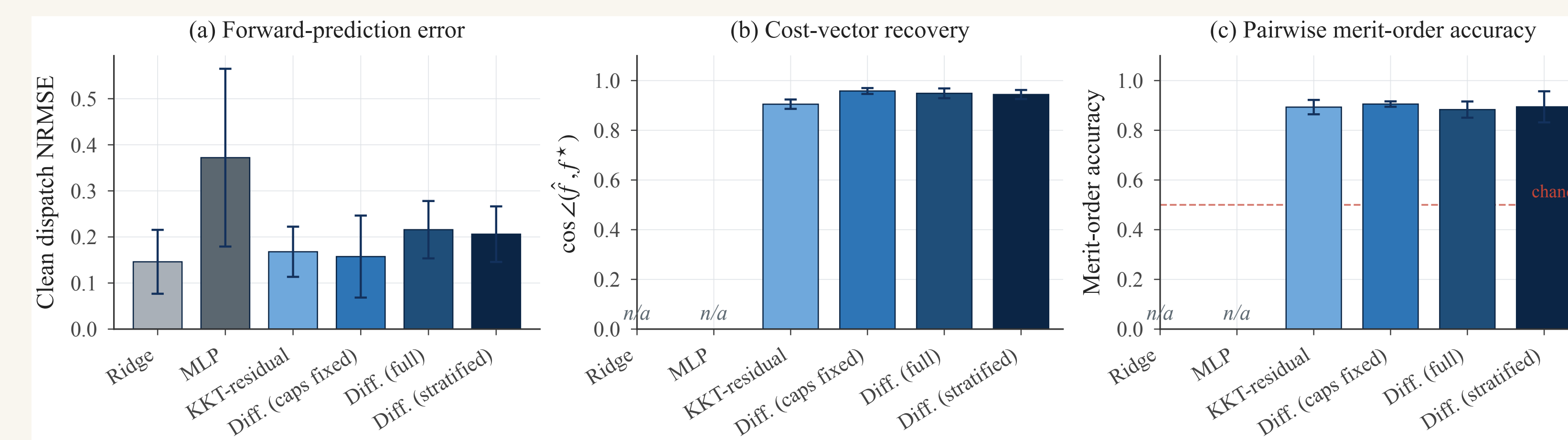
Three extensions over the F&D baseline:

- 1 Extended parameters.** Jointly learn $(f, g_{\max}, p_{\max})$ — not just costs.
- 2 Stratified costs.** Laplacian-regularized per-stratum $f^{(s)}$ captures diurnal merit-order swing.
- 3 Robust loss.** Huber residual tolerates sensor noise, derates, and unit-commitment effects.

Headline Results — 10-Bus Synthetic Benchmark

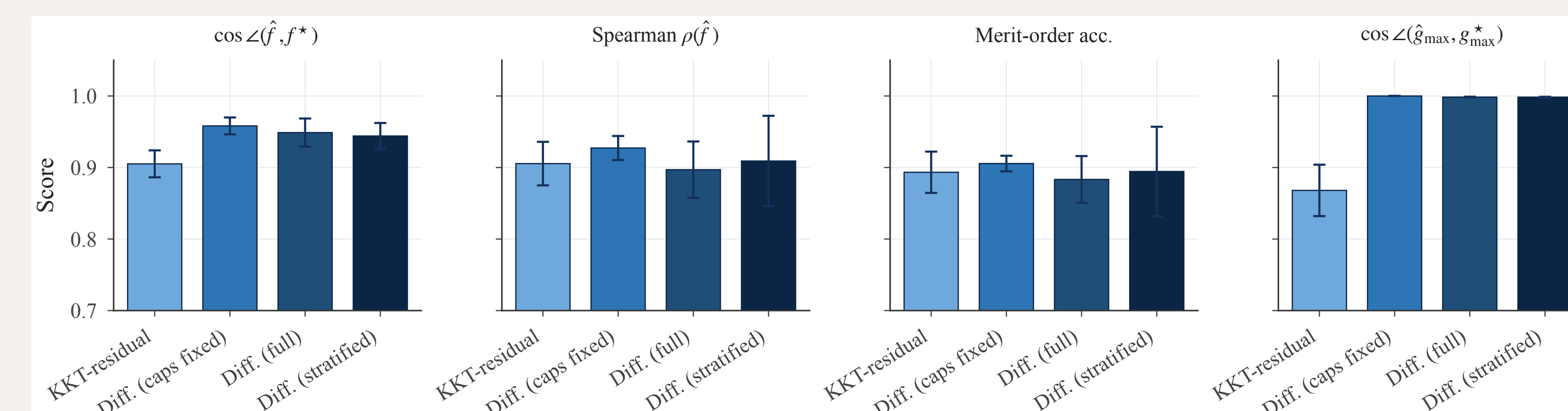
3 seeds, mean $\pm$ std. *Diff.* rows = this work. Clean NRMSE: forward error on noise-free GT dispatch.

Method	NRMSE _{cl.}	$\cos \hat{f}$	Merit acc.	$\cos \hat{g}_{\max}$
Ridge	0.110	—	—	—
MLP	0.477	—	—	—
KKT baseline	0.138	.908 $\pm$.019	.881 $\pm$.034	.870 $\pm$.055
Diff. (caps fixed)	0.125	.959$\pm$.012	.904$\pm$.013	1.00
Diff. (full)	0.208	.949 $\pm$.018	.889 $\pm$.022	.999
Diff. (strat.)	0.198	.947 $\pm$.018	.874 $\pm$.046	.999

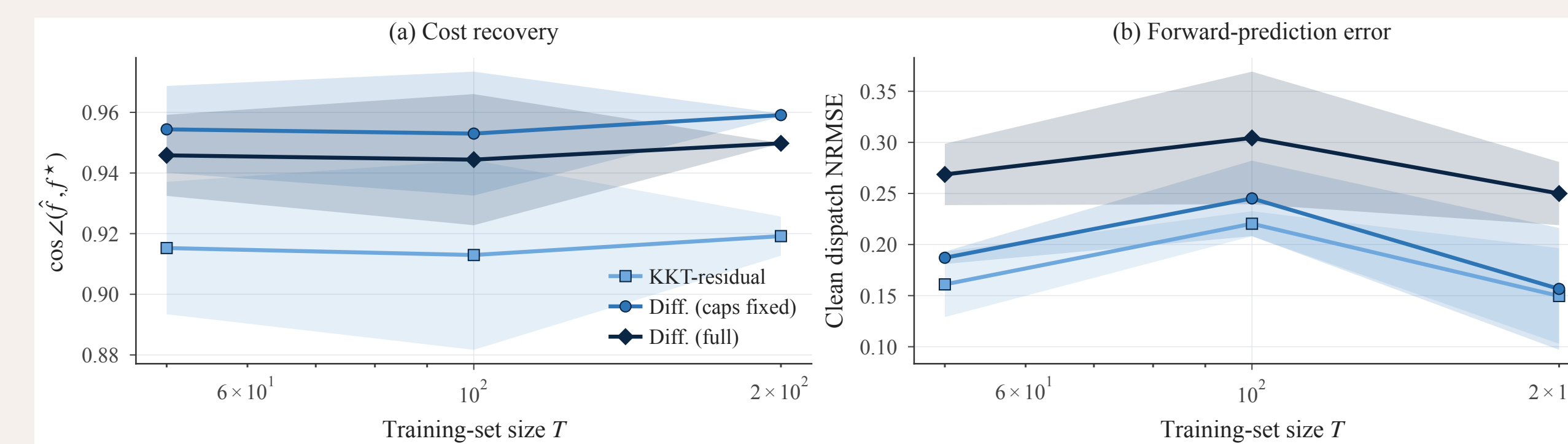


(a) Clean dispatch NRMSE. (b) Cosine recovery of $\hat{f}$. (c) Merit-order accuracy. Regression baselines (Ridge, MLP) overfit noise and recover *no* structural parameters.

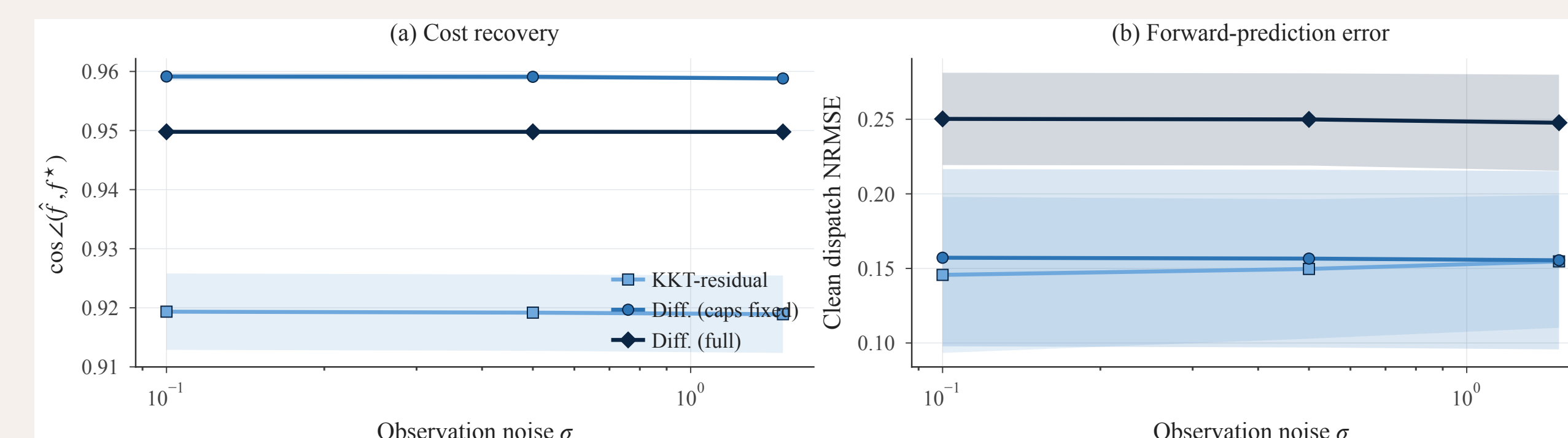
Parameter Recovery Detail



Sample-Size & Noise Robustness

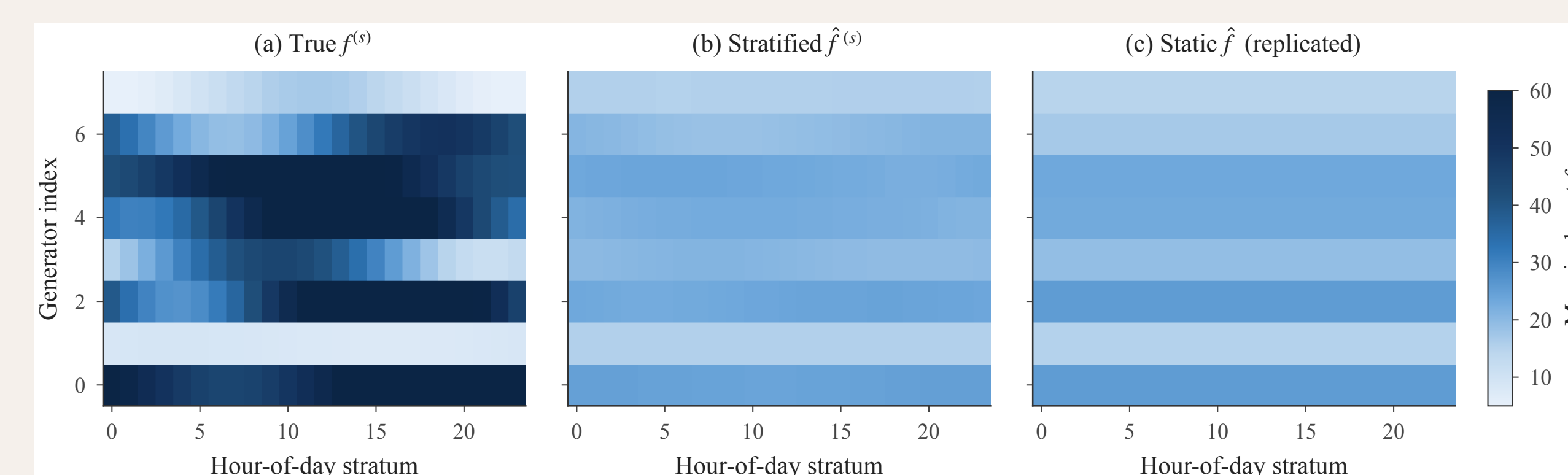


Recovery vs. training size $T \in \{50, 100, 200, 400\}$ (3 seeds; bands = $\pm 1\sigma$). Differentiable estimator dominates KKT at every T .



Recovery vs. observation noise σ (3 seeds). Gap widens under higher noise where the Huber loss provides extra robustness.

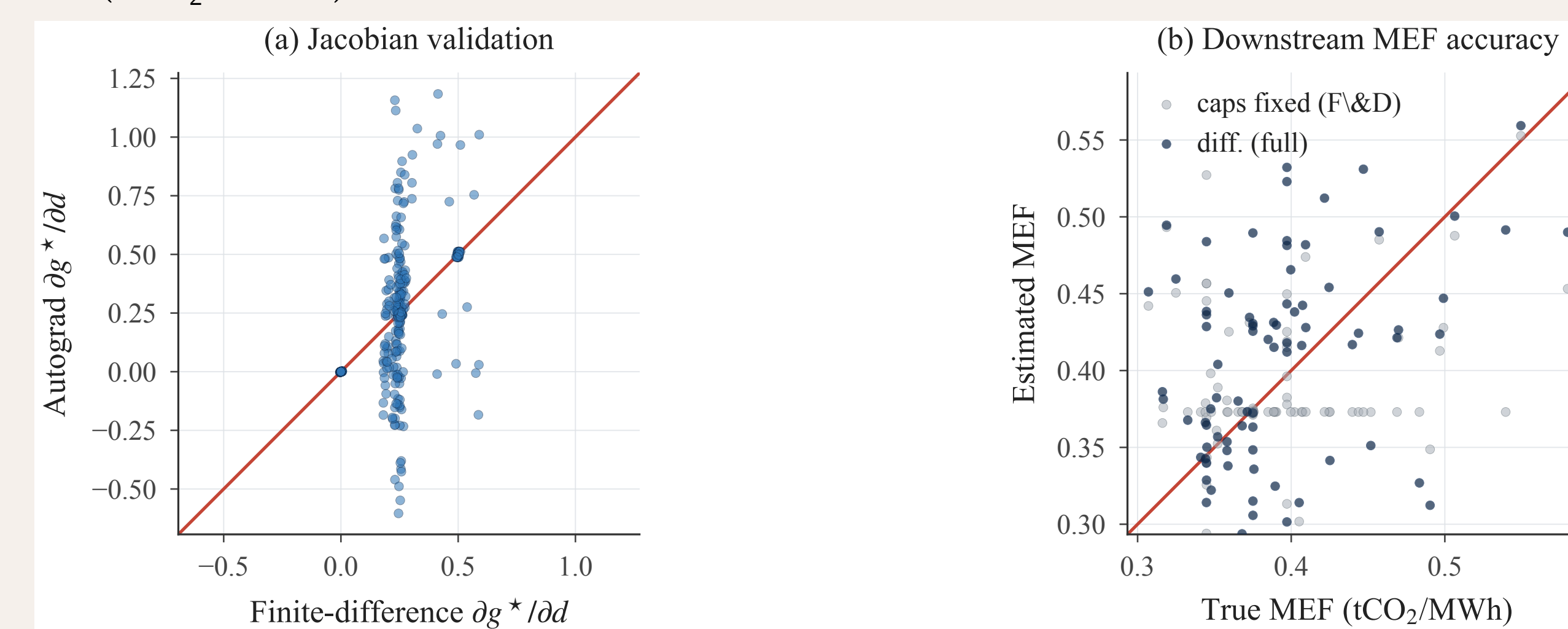
Diurnal Cost Recovery



Hour-of-day cost table for 8-bus network with 60% diurnal cost amplitude. **Truth** (left) vs. **Stratified** (centre) vs. **Static** (right). The static model collapses the diurnal swing; the stratified estimator recovers it faithfully.

Downstream: Marginal Emission Factors

Autograd Jacobian $\partial g^* / \partial d$ enables exact MEFs: $\text{MEF}(d) = \nabla_d (c_{CO_2}^\top g^*(d))$.



(a) Autograd vs. finite-difference Jacobian validation. (b) Estimated vs. true MEFs; both full and caps-fixed estimators achieve **cosine 0.98** (RMSE = 0.10).

Conclusions & Contributions

► **Joint recovery** of $\hat{f}, \hat{g}_{\max}, \hat{p}_{\max}$ with **cosine 0.96** merit-order accuracy and **0.99**